



The
University
Of
Sheffield.

School of Electrical and
Electronic Engineering

Project and Dissertation – Progress Review Form

It is the student's responsibility to complete this form

The student should discuss their progress with their supervisor when completing the form

The completed form should be included in the student's submitted dissertation (in the Appendix)

Student name: Dharani Saravanan

Student Reg No: 250120921

Project code and Title: ELE468: Project and Dissertation

Supervisor: Shuhei Miyashita

Second Reader: To be allocated

Project Progress against objectives (in bullet-points):

Basic Objectives:

1. O-1: Develop an ML-assisted evolutionary approach using Python to generate and evolve two-dimensional self-assembling module configurations, with the specific ML approach to be selected following a review of relevant literature. - **To be done**
2. O-2: Extend the existing simulation framework to enable self-folding of two-dimensional module configurations into three-dimensional robotic structures. - **Done**
3. O-3: Extend the existing simulation framework to evaluate three-dimensional robotic structures within a physics-based simulation environment and actuate them using an oscillating magnetic field to achieve locomotion performance. Feed the resulting fitness values back into the evolutionary framework. - **In Progress**
4. O-4: Analyse the evolution of robotic morphologies and investigate their influence on locomotion behaviour and performance using performance-based metrics across generations - **To be done**

Advanced Objectives:

1. AO-1: Investigate and compare different ML surrogate-learning approaches for morphology fitness prediction and evaluate their influence on optimisation efficiency and evolutionary performance. - **To be done**
2. AO-2: Investigate entropy and morphology-complexity measures and analyse their relationship with locomotion behaviour and evolutionary outcomes. - **To be done**

Main tasks to be completed in the next 4 weeks (in bullet-points):

- *Week1 (JUL 6 - 12) - Finish Simulation for actuation*
 - *Task in Gantt chart: Simulation for self-folding & Actuation*
- *Week2 (JUL 13 - JUL 19) - Formulize fitness function*
 - *Task in Gantt chart: Design and develop evolutionary framework*
- *Week3 (JUL 20 - JUL 27) - LIT review for Evolutionary framework and ML*
 - *Task in Gantt chart: Literature review and background research*
- *Week4 (JUL 28 - AUG2) - Acquire Evolutionary results*
 - *Task in Gantt chart: Design and develop evolutionary framework*
- *Week5 (AUG3 - AUG8) - Train ML model for prediction*
 - *Task in Gantt chart: Develop ML-assisted optimisation*

Please attach (as an additional sheet) your updated project work plan in the form of a Gantt chart

Student name (Print): DHARANI SARAVANAN

Date: 06-07-2026

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